INCH-POUND

MIL-B-29597(NAVY)
6 April 1993

MILITARY SPECIFICATION

BAG, ODOR BARRIER, FLEXIBLE (FOR FOOD CONTAMINATED PLASTIC WASTE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of an odor barrier bag to be used to contain food-contaminated plastic waste (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center, Aircraft Division, Systems Requirements Department (Code SR32), Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8105

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection
MIL-STD-129	-	Marking for Shipment and Storage.

(Unless otherwise specified, copies of federal and military specifications and standards are available from the DODSSP - Customer Service, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 882	-	Tensile Properties of Thin Plastic Sheeting
ASTM D 1922	-	Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method
ASTM D 3985	-	Test Method for Oxygen Gas Transmission Rate Through Plastic Film Sheeting Using a Coulometric Sensor
ASTM F 88	-	Test Method for Seal Strength of Flexible Barrier Materials
ASTM F 1249	-	Standard Test Method for Water - Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 4.3.1, 6.2, and 6.3), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. The film shall be made from such material and by such a process as to ensure compliance with the requirements of this specification.

3.3 Form. The film shall be furnished in rolls formed of lay flat tubing.

3.3.1 Rolls. The average roll length shall be 75 ± 0.5 yards (64 ± 0.5 meters). The length of any individual roll shall not be less than 67 yards (61 meters). The width of roll material shall be 36 ± 0.25 inches (91 ± 0.6 cm). The length of the core shall be equal to the width of roll material, with a plus tolerance of 0.25 inches (0.64 cm). The core shall be of sufficient rigidity to prevent distortion of the roll under normal conditions during transportation, storage and use. Each roll shall be suitably restrained to prevent unwinding.

3.3.1.1 Bags. Bags shall be created on the roll of lay flat tubing, by creating a heat seal every 50 + 0.25 inches (127 + 0.64 cm) on the roll. Heat seals shall be of high integrity (see 3.9). Perforations shall be created to enable bags to be removed from the roll by hand. The perforations shall occur below the heat seal, and shall not be more than 0.25 inches (0.64 cm) below the effective heat seal. Perforations shall not interfere with the integrity of the seal.

3.4 Thickness. Unless otherwise specified (see 6.2), the film thickness shall be 0.005 inches (0.013 cm). When other thicknesses are desired, they shall be specified in increments of 0.001 inches (0.025 cm). The tolerance shall be ± 5 percent of the specified thickness when measuring the average gage.

3.5 Solvent Compatibility. The film shall be capable of meeting 75 percent of each mechanical property value specified in 3.7 and shall reveal no adverse visual defects after treatment in accordance with 4.5.1 with any of the following solvents:

a. Chloroform
b. Xylene
c. Kerosene

3.6 Permeability. The water vapor permeability shall not exceed 5.97 grams per square meter per day for 0.005 inch (0.013 cm) thick film material when tested in accordance with 4.5.2. The oxygen permeability shall not exceed 1.16 cubic centimeters (standard temperature and pressure) per square meter per day per atmosphere for 0.005 inches (0.013 cm) thick film material, when tested in accordance with 4.5.2.

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3.7 Tensile Strength, Tear Strength and Elongation. The film tensile strength shall be at least 4910 pounds per square inch (33,800 megapascals) in the machine direction and 4700 pounds per square inch (32,400 megapascals) in the transverse direction, when tested in accordance with 4.5.3. The film elongation shall be at least 280 percent in the machine and 294 percent in the transverse direction, when tested in accordance with 4.5.3. The film tear strength shall be at least 96 grams in the machine direction and 124 grams in the transverse direction, when tested in accordance with 4.5.3.

3.8 Identification Sheet. An identification sheet shall accompany each roll (see 5.1) and shall contain the specification number, manufacturer's name, manufacturer's designation, lot number, and date of manufacture (month and year). The sheet shall provide the following recommended sealing information (see 3.9): type of equipment, temperature, pressure and dwell time.

3.9 Heat Seal Requirements. Heat seals shall be 0.25 ± 0.05 inches (0.64 ± 0.13 cm). Seals shall be the flat impulse type whereby the film is heated to its welding or melting temperature under pressure and allowed to cool while still under pressure. Heat seals shall be of high integrity with a minimum breaking strength of 15 pounds per inch (263 newtons per meter), when tested in accordance with 4.5.4.

3.10 Workmanship. The film shall be manufactured in a manner to provide uniform construction, free from pinholes, tears, cuts, sharp creases, or other imperfections which might impair its usefulness for the purpose intended. The film shall be trimmed of any selvage and the finished product shall conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of all the tests specified in section 3 and inspections of this specification. The responsibility for the performance of the first article inspection shall be as specified in the contract or purchase order (see 4.3.1, 6.2, 6.3, and 6.4). Approval of the first article inspection sample does not preclude the requirements for performing the quality conformance inspection.

4.3.1 Prior approval. If a contractor has previously delivered an acceptable product meeting the requirements of this specification, first article inspection may be waived at the discretion of the procuring activity for a period of time not to exceed 2 years (see 6.2).

4.3.2 First article sample. Unless otherwise specified by the procuring activity (see 6.2), the contractor shall submit one roll of film, 70 yards by 36 inches (64 m by 91 cm). The first article sample shall be produced using the same plant facilities intended for the production run. Failure of the first article sample to meet all the requirements of this specification shall be cause for rejection.

4.3.2.1 Data to be submitted with first article samples. The supplier shall submit two copies of the first article inspection report showing conformance to all of the test requirements contained in section 3 (see 6.2 and 6.3).

4.3.3 Certified statement. Whether or not first article inspection is required (see 4.3.1), the contractor shall certify in writing that the material meets all the requirements of this specification (see 6.2 and 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of all the inspections and tests listed in 4.4.2.

4.4.1 Sampling.

4.4.1.1 Lot size. An inspection lot shall consist of 665 rolls of material, manufactured by the same process, from the same components, by one manufacturer, at one plant, and offered for delivery at one time.

4.4.1.2 Sampling for inspection. Samples for examination and testing shall be selected in accordance with inspection level III, table II, single sampling, tightened inspection of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.0 percent defective.

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4.4.2 Quality conformance.

4.4.2.1 Quality conformance tests.

PROPERTY	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Solvent compatibility	3.5	4.5.1
Permeability	3.6	4.5.2
Tensile strength	3.7	4.5.3
Tear strength	3.7	4.5.3
Elongation	3.7	4.5.3
Seal strength	3.9	4.5.4

4.4.2.2 Inspection of end item for defects.

EXAMINE	DEFECTS
Form	Rolled or flat cut not as specified, lay flat dimensions or film thickness not as specified
Cleanliness	Not clean
Workmanship	Embrittlement or any holes, tears, cuts chafed spots, or other defects that would impair usefulness of film
Construction	Not uniform; pinholes, cracks, blister scratches, mottling, folds, foreign matters, gels, or fish eyes
Color	Not uniform
Identification	Marking not as specified; identification sheet missing

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4.4.2.3 Inspection of packaging for defects.

EXAMINE	DEFECTS
Preservation	Not in accordance with contract requirements
Packing	Not in accordance with contract requirements
Markings	Markings illegible, incorrect omitted

4.5 Test methods. All tests shall be conducted as specified herein. Unless otherwise specified, prior to testing, film samples shall be conditioned for 24 hours (min) in a circulating air environment having a relative humidity of 50 ± 5 percent and a temperature of 70° to 76°F (21° to 24.6°C).

4.5.1 Solvent compatibility. Film samples shall be immersed in the solvents (a separate specimen for each solvent) specified in 3.5 for 7 days at 68° ± 5°F (20° ± 3°C). Visual examination after treatment shall reveal no adverse defects. The treated film shall be tested in accordance with 4.5.3 to determine compliance with 3.5.

4.5.2 Permeability. The water-vapor permeability shall be measured in accordance with ASTM F 1249. The oxygen permeability shall be tested in accordance with ASTM D 3985. Oxygen permeability shall be tested at 0% relative humidity.

4.5.3 Film tensile strength, tear strength, and elongation. The film tensile strength and elongation shall be tested in accordance with ASTM D 882. The film tear strength shall be tested in accordance with ASTM D 1922.

4.5.4 Seal strength. The film seal strength shall be tested in accordance with ASTM F 88.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired levels of protection shall be as specified by the procuring activity (see 6.2). Additionally, the identification sheet (see 3.8) shall be inserted with each roll.

5.2 Marking. All individual packs and shipping containers shall be marked in accordance with MIL-STD-129 and with the following: specification number, lot number, size - nominal net linear yardage of roll, date of material manufacture (month, year), and the statement "Keep Cool and Dry."

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Flexible odor barrier bags in accordance with this specification are intended to be used for the containment of food-contaminated plastic waste in shipboard applications where at-sea disposal of plastics is prohibited.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Thickness (if other than 0.005 inches) (see 3.4)
d. Details for first article inspections/testing (see 4.3-4.3.3 and 6.4).
e. Packaging requirements and levels of protection (see 5.1)

6.2.1 Part of identifying number (PIN). Military part numbers should be coded in accordance with the following example:

EXAMPLE:

M29597

└——— Denotes basic specification MIL-B-29597(Navy)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID is tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Para.	DID Number	DID Title	Suggested Tailoring
4.3.2.1	DI-NDTI-80809A	TEST/INSPECTION REPORTS	---
4.3.3	"	"	10.2.7.2

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.4 First article. The contracting officer should provide specific guidance to offerors whether the first article samples should be a preproduction item, a first production item, a sample selected from the first production items or a standard production item from the contractor's current inventory (see 3.1 and 4.3). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for submission of Material Safety Data Sheets (MSDS) prior to contract award, testing, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive requirements for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.8 Subject term (keyword) listing.

Food contamination
Odor barrier
Plastic film
Plastic waste
Waste bag

Custodian:
Navy - AS

Preparing Activity:
Navy - AS
(Project 8135-N359)

Review Activity:
Navy - SA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-29597 (NAVY)	6 April 1993

3. DOCUMENT TITLE
BAG, ODOR BARRIER, FLEXIBLE (FOR FOOD CONTAMINATED PLASTIC WASTE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. [redacted]MMECOMMANDING OFFICER, NAVAL AIR [redacted]RFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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